

SIMATS ENGINEERING

ASSESSMENT TOOL 2 (LOW) - Industry Problem Solving Task

COURSE NAME: R PROGRAMMING

COURSE CODE: ITA04

COURSE OUTCOME COVERED

CO3 Data Frames, Making data frames - Working with data frames- Data Reshaping- Melting and Casting of data – Merging Data Frames - Editing and Reading Data from Files –Reading and Writing Files.

(Bloom Level -BL4- BL5)

Assessment Tool Weightage: 25%

Code of Conduct:

I Adhithyan S (192421102) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS:

Total Marks: 100

1. Healthcare Analytics

A hospital maintains two datasets: Patient.csv (PatientID, Name, Age, Disease) and Treatment.csv (PatientID, Doctor, TreatmentCost). Read both datasets and merge them to generate a report showing each patient's treatment details. Export the merged report as Hospital_Report.csv.

2. Retail Sales Analysis

A supermarket stores monthly sales data in separate CSV files for different branches. Read all datasets into R, merge them into a single data frame, and calculate the total sales for each branch. Save the consolidated report to a new CSV file.

3. Manufacturing Quality Control

A manufacturing company records product defects in separate production-line files. Combine the datasets, reshape the data to summarize defect counts by product category, and identify the production line with the highest defect rate.

4. Banking Customer Analysis

A bank maintains customer information in one file and loan details in another. Merge both datasets using the Customer ID and generate a report showing customers with loans exceeding ₹10,00,000. Save the filtered report.

5. E-Commerce Product Analysis

An online shopping company stores product details and customer reviews separately. Read both datasets, merge them, and reshape the data to calculate the average product rating for each product category.

6. Human Resource Management

An organization maintains Employee.csv and Salary.csv datasets. Merge the datasets, create a new column called Annual Salary, update missing salary values, and export the updated employee database.

7. Smart Agriculture Monitoring

A smart farming system records daily temperature, humidity, and crop yield. Reshape the dataset from wide format to long format using melt(), analyze environmental parameters, and reconstruct the original dataset using dcast().

8. Logistics and Supply Chain

A logistics company records shipment details in one dataset and delivery status in another. Merge both datasets to identify delayed deliveries and generate a city-wise shipment summary.

9. University Student Performance System

A university stores student details and semester marks in separate files. Read both datasets, merge them using Student ID, calculate each student's average marks, and export the final result sheet.

10. Insurance Claim Processing

An insurance company maintains policy details and claim records separately. Merge the datasets to identify customers with multiple claims, reshape the data to summarize claims by policy type, and save the processed report.

11. Telecommunications Network Monitoring

A telecom company records daily network performance metrics in multiple files. Combine all datasets, reshape the data to compare download speed across regions, and generate a performance report for management.

12. Energy Consumption Analysis

A smart electricity board collects hourly electricity consumption data from different regions. Convert the dataset from wide format to long format using `melt()`, summarize daily consumption using `dcast()`, and export the analysis.

13. Airline Reservation System

An airline maintains passenger booking details and flight schedules in separate datasets. Merge both datasets using Flight Number, identify delayed flights, update missing records, and save the final report.

14. Financial Market Analysis

A financial organization stores daily stock prices and company information separately. Read the datasets, merge them, reshape the stock prices to compare company-wise monthly performance, and export the summarized report.

15. Smart City Traffic Management

A smart city traffic monitoring system collects traffic volume from multiple intersections every hour. Read all traffic datasets, merge them, reshape the data to calculate peak-hour traffic by location, and generate a CSV report for urban planners.

ANSWERS

1. HEALTHCARE ANALYTICS

Input:

```
patient <- data.frame(

  PatientID = c(101, 102, 103, 104, 105),

  Name = c("Arun", "Bala", "Charan", "Divya", "Esha"),

  Age = c(45, 32, 56, 28, 40),

  Disease = c("Diabetes", "Fever", "Hypertension", "Asthma", "Diabetes")

)

print(patient)

treatment <- data.frame(

  PatientID = c(101, 102, 103, 104, 105),

  Doctor = c("Dr. Kumar", "Dr. Ravi", "Dr. Meena",

             "Dr. Anitha", "Dr. Kumar"),

  TreatmentCost = c(5000, 2500, 7000, 4000, 5500)

)

print(treatment)

hospital_report <- merge(patient, treatment, by = "PatientID")

print("Hospital Treatment Report:")
```

```

print(hospital_report)

write.csv(hospital_report,

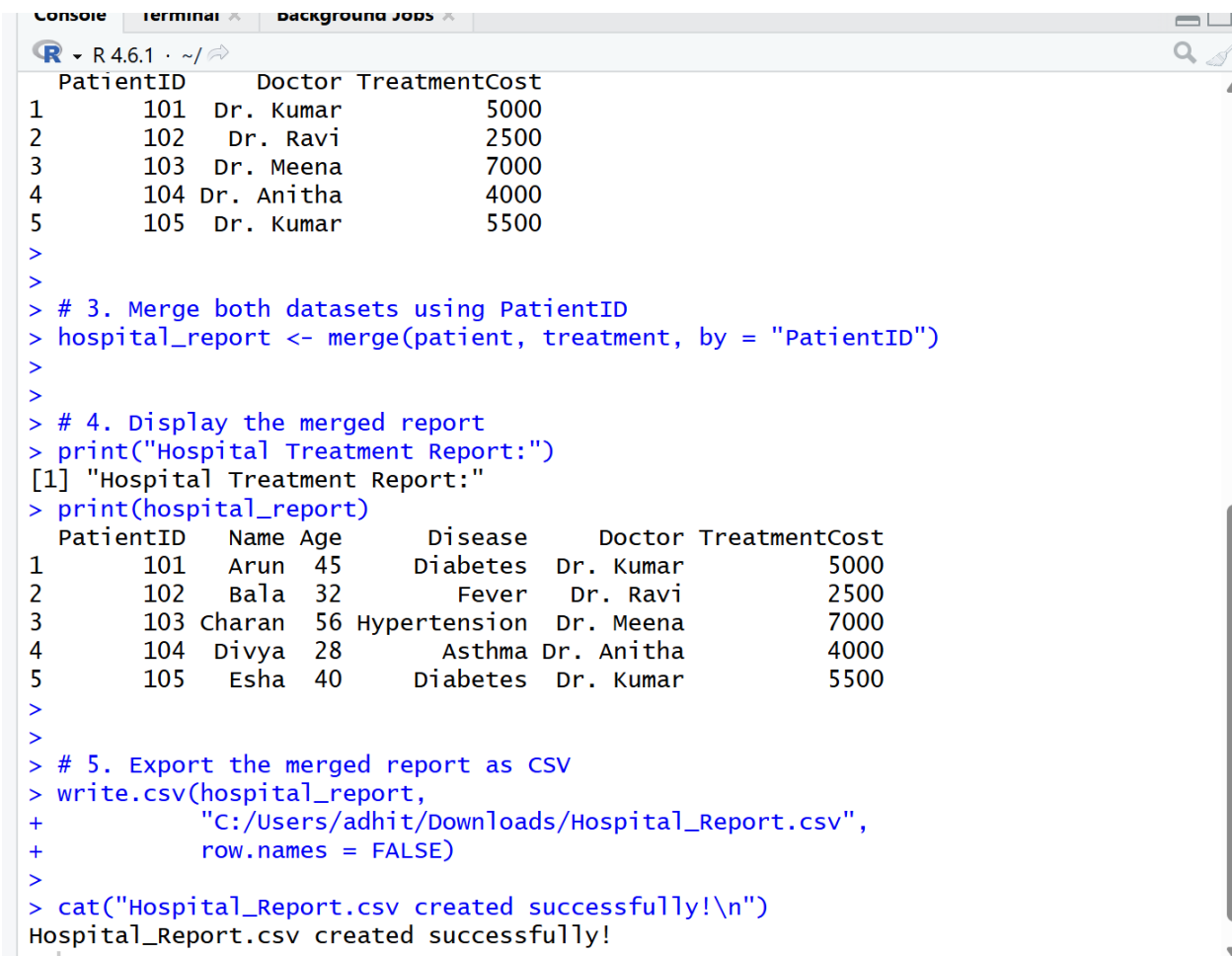
         "C:/Users/adhit/Downloads/Hospital_Report.csv",

         row.names = FALSE)

cat("Hospital_Report.csv created successfully!\n")

```

Output:



```

R 4.6.1 ~ /
PatientID Doctor TreatmentCost
1 101 Dr. Kumar 5000
2 102 Dr. Ravi 2500
3 103 Dr. Meena 7000
4 104 Dr. Anitha 4000
5 105 Dr. Kumar 5500
>
>
> # 3. Merge both datasets using PatientID
> hospital_report <- merge(patient, treatment, by = "PatientID")
>
>
> # 4. Display the merged report
> print("Hospital Treatment Report:")
[1] "Hospital Treatment Report:"
> print(hospital_report)
PatientID Name Age Disease Doctor TreatmentCost
1 101 Arun 45 Diabetes Dr. Kumar 5000
2 102 Bala 32 Fever Dr. Ravi 2500
3 103 Charan 56 Hypertension Dr. Meena 7000
4 104 Divya 28 Asthma Dr. Anitha 4000
5 105 Esha 40 Diabetes Dr. Kumar 5500
>
>
> # 5. Export the merged report as CSV
> write.csv(hospital_report,
+           "C:/Users/adhit/Downloads/Hospital_Report.csv",
+           row.names = FALSE)
>
> cat("Hospital_Report.csv created successfully!\n")
Hospital_Report.csv created successfully!

```

2. RETAIL SALES ANALYSIS

Input:

```
branch1 <- data.frame(

  Month = c("January", "February", "March"),

  Sales = c(25000, 28000, 30000)

)

branch2 <- data.frame(

  Month = c("January", "February", "March"),

  Sales = c(22000, 26000, 29000)

)

branch3 <- data.frame(

  Month = c("January", "February", "March"),

  Sales = c(30000, 32000, 35000)

)

write.csv(branch1, "Branch1.csv", row.names = FALSE)

write.csv(branch2, "Branch2.csv", row.names = FALSE)

write.csv(branch3, "Branch3.csv", row.names = FALSE)

branch1 <- read.csv("Branch1.csv")

branch2 <- read.csv("Branch2.csv")
```

```
branch3 <- read.csv("Branch3.csv")

branch1$Branch <- "Branch 1"

branch2$Branch <- "Branch 2"

branch3$Branch <- "Branch 3"

sales_data <- rbind(branch1, branch2, branch3)

print("Consolidated Sales Data:")

print(sales_data)

total_sales <- aggregate(

  Sales ~ Branch,

  data = sales_data,

  FUN = sum

)

print("Total Sales for Each Branch:")

print(total_sales)

write.csv(

  total_sales,

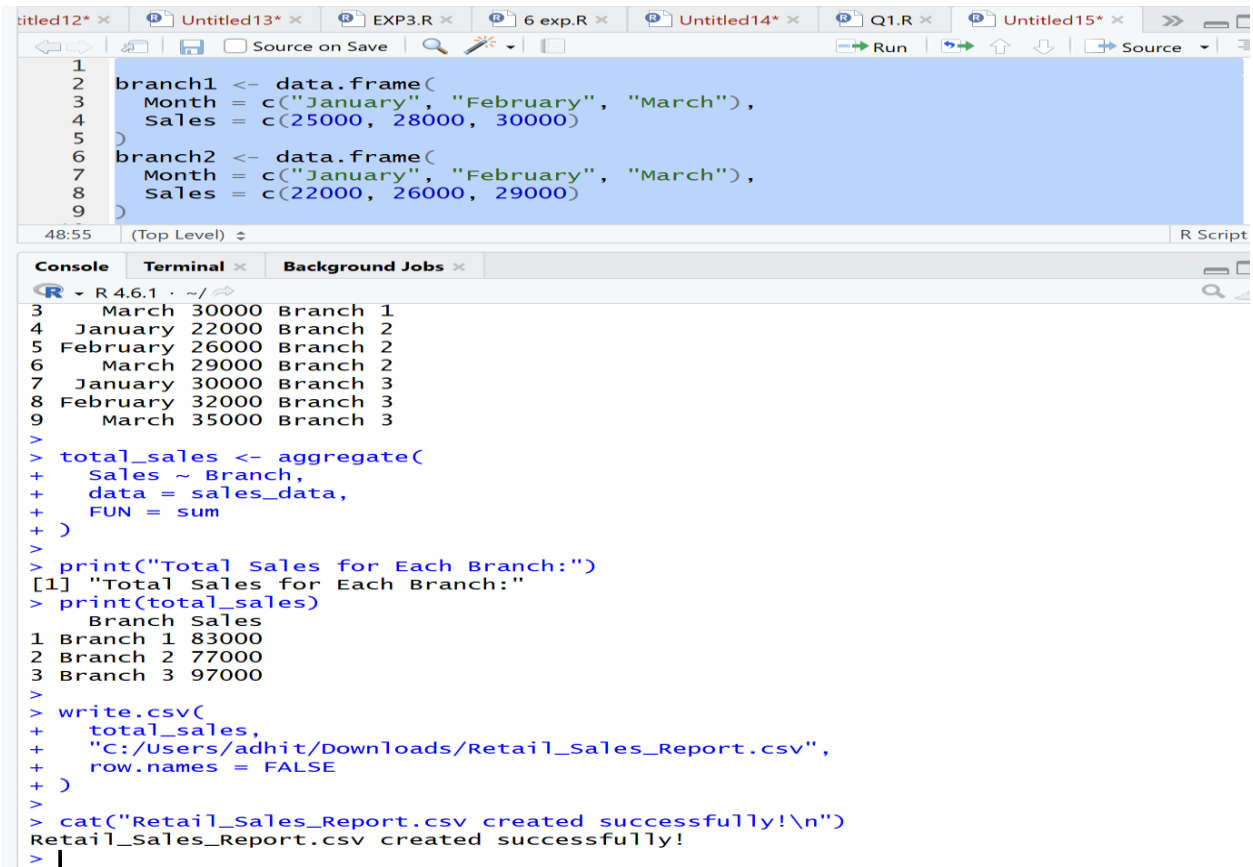
  "C:/Users/adhit/Downloads/Retail_Sales_Report.csv",

  row.names = FALSE

)
```

```
cat("Retail_Sales_Report.csv created successfully!\n")
```

Output:



The screenshot shows the R Studio environment. The top pane displays the source code for an R script. The bottom pane shows the console output, which includes the execution of the code and the resulting data structures.

```
1 branch1 <- data.frame(  
2   Month = c("January", "February", "March"),  
3   Sales = c(25000, 28000, 30000)  
4 )  
5  
6 branch2 <- data.frame(  
7   Month = c("January", "February", "March"),  
8   Sales = c(22000, 26000, 29000)  
9 )  
  
> total_sales <- aggregate(  
+   Sales ~ Branch,  
+   data = sales_data,  
+   FUN = sum  
+ )  
> print("Total Sales for Each Branch:")  
[1] "Total Sales for Each Branch:"  
> print(total_sales)  
      Branch Sales  
1 Branch 1 83000  
2 Branch 2 77000  
3 Branch 3 97000  
> write.csv(  
+   total_sales,  
+   "C:/Users/adhit/Downloads/Retail_Sales_Report.csv",  
+   row.names = FALSE  
+ )  
> cat("Retail_Sales_Report.csv created successfully!\n")  
Retail_Sales_Report.csv created successfully!  
> |
```

3. Manufacturing Quality Control

Input:

```
line1 <- data.frame(  
  
  ProductionLine = "Line 1",  
  
  ProductCategory = c("Electronics", "Furniture", "Clothing",  
  
    "Electronics", "Furniture"),  
  
  TotalProducts = c(100, 120, 150, 110, 130),
```

```

    Defects = c(5, 8, 6, 4, 7)

)

line2 <- data.frame(

    ProductionLine = "Line 2",

    ProductCategory = c("Electronics", "Furniture", "Clothing",

        "Electronics", "Furniture"),

    TotalProducts = c(110, 100, 140, 120, 125),

    Defects = c(8, 5, 10, 7, 9)

)

line3 <- data.frame(

    ProductionLine = "Line 3",

    ProductCategory = c("Electronics", "Furniture", "Clothing",

        "Electronics", "Furniture"),

    TotalProducts = c(105, 115, 130, 125, 140),

    Defects = c(3, 6, 5, 4, 8)

)

write.csv(line1, "Production_Line1.csv", row.names = FALSE)

write.csv(line2, "Production_Line2.csv", row.names = FALSE)

```

```
write.csv(line3, "Production_Line3.csv", row.names = FALSE)
```

```
line1 <- read.csv("Production_Line1.csv")
```

```
line2 <- read.csv("Production_Line2.csv")
```

```
line3 <- read.csv("Production_Line3.csv")
```

```
quality_data <- rbind(line1, line2, line3)
```

```
print("Combined Manufacturing Quality Data:")
```

```
print(quality_data)
```

```
defect_summary <- aggregate(
```

```
  Defects ~ ProductCategory,
```

```
  data = quality_data,
```

```
  FUN = sum
```

```
)
```

```
print("Defect Counts by Product Category:")
```

```
print(defect_summary)
```

```
line_summary <- aggregate(
```

```
  cbind(TotalProducts, Defects) ~ ProductionLine,
```

```
  data = quality_data,
```

```
  FUN = sum
```

```
)
```

```

line_summary$DefectRate <-

(line_summary$Defects / line_summary$TotalProducts) * 100

print("Production Line Defect Rates:")

print(line_summary)

highest_line <- line_summary[

  which.max(line_summary$DefectRate),

]

print("Production Line with Highest Defect Rate:")

print(highest_line)

write.csv(

  line_summary,

  "C:/Users/adhit/Downloads/Manufacturing_Quality_Report.csv",

  row.names = FALSE

)

cat("Manufacturing_Quality_Report.csv created successfully!\n")

```

Output:

```
30 line3 <- read.csv("Production_Line3.csv")
31
32 quality_data <- rbind(line1, line2, line3)
33
34 print("Combined Manufacturing Quality Data:")
35 print(quality_data)
36
37 defect_summary <- aggregate(
38   Defects ~ ProductCategory,
39   data = quality_data,
40   FUN = sum)
41
42 line_summary <- data.frame(
43   ProductionLine = 1:3,
44   TotalProducts = sum(quality_data$TotalProducts, na.rm=T),
45   Defects = sum(quality_data$Defects, na.rm=T),
46   DefectRate = (sum(quality_data$Defects, na.rm=T) / sum(quality_data$TotalProducts, na.rm=T)) * 100
47 )
48
49 print("Production Line Defect Rates:")
50 [1] "Production Line Defect Rates:"
51 print(line_summary)
52
53   ProductionLine TotalProducts Defects DefectRate
54 1           Line 1           610       30    4.918033
55 2           Line 2           595       39    6.554622
56 3           Line 3           615       26    4.227642
57
58 highest_line <- line_summary[
59   which.max(line_summary$DefectRate),
60 ]
61
62 print("Production Line with Highest Defect Rate:")
63 [1] "Production Line with Highest Defect Rate:"
64 print(highest_line)
65
66   ProductionLine TotalProducts Defects DefectRate
67 2           Line 2           595       39    6.554622
68
69 write.csv(
70   line_summary,
71   "C:/Users/adhit/Downloads/Manufacturing_Quality_Report.csv",
72   row.names = FALSE
73 )
74
75 cat("Manufacturing_Quality_Report.csv created successfully!\n")
76 Manufacturing_Quality_Report.csv created successfully!
```

4. BANKING CUSTOMER ANALYSIS

Input:

```
customer <- data.frame(
```

```
  CustomerID = c(101, 102, 103, 104, 105),
```

```
  Name = c("Arun", "Bala", "Charan", "Divya", "Esha"),
```

```
  Age = c(35, 42, 29, 51, 38),
```

```
  City = c("Chennai", "Coimbatore", "Madurai", "Salem", "Trichy")
```

```
)
```

```
loan <- data.frame(
```

```
  CustomerID = c(101, 102, 103, 104, 105),
```

```
  LoanType = c("Home Loan", "Car Loan", "Home Loan", "Business Loan", "Home Loan"),
```

```

LoanAmount = c(1500000, 750000, 1200000, 2000000, 900000)

)

write.csv(customer, "Customer.csv", row.names = FALSE)

write.csv(loan, "Loan.csv", row.names = FALSE)

customer <- read.csv("Customer.csv")

loan <- read.csv("Loan.csv")

bank_report <- merge(customer, loan, by = "CustomerID")

print("Merged Banking Customer Report:")

print(bank_report)

filtered_report <- subset(bank_report, LoanAmount > 1000000)

print("Customers with Loans Exceeding Rs. 10,00,000:")

print(filtered_report)

write.csv(

    filtered_report,

    "C:/Users/adhit/Downloads/Banking_Loan_Report.csv",

    row.names = FALSE

)

cat("Banking_Loan_Report.csv created successfully!\n")

```

Output:

```

28 print(filtered_report)
29
30 write.csv(
31   filtered_report,
32   "C:/Users/adhit/Downloads/Banking_Loan_Report.csv",
33   row.names = FALSE
34 )
35
36

```

36:55 (Top Level) R Sc

Console Terminal Background Jobs

R 4.6.1 ~ /

```

> bank_report <- merge(customer, loan, by = "CustomerID")
>
> print("Merged Banking Customer Report:")
[1] "Merged Banking Customer Report:"
> print(bank_report)
  CustomerID  Name Age   City      LoanType LoanAmount
1         101  Arun  35  Chennai      Home Loan    1500000
2         102  Bala  42 Coimbatore    Car Loan     750000
3         103 Charan 29   Madurai    Home Loan    1200000
4         104 Divya 51   Salem Business Loan    2000000
5         105  Esha  38    Trichy      Home Loan     900000
>
> filtered_report <- subset(bank_report, LoanAmount > 1000000)
>
> print("Customers with Loans Exceeding Rs. 10,00,000:")
[1] "Customers with Loans Exceeding Rs. 10,00,000:"
> print(filtered_report)
  CustomerID  Name Age   City      LoanType LoanAmount
1         101  Arun  35  Chennai      Home Loan    1500000
3         103 Charan 29   Madurai    Home Loan    1200000
4         104 Divya 51   Salem Business Loan    2000000
>
> write.csv(
+   filtered_report,
+   "C:/Users/adhit/Downloads/Banking_Loan_Report.csv",
+   row.names = FALSE
+ )
>
> cat("Banking_Loan_Report.csv created successfully!\n")
Banking_Loan_Report.csv created successfully!
>

```

5. E-COMMERCE PRODUCT ANALYSIS

Input:

```
product <- data.frame(
```

```
  ProductID = c(101, 102, 103, 104, 105),
```

```
  ProductName = c("Laptop", "Phone", "Headphones", "Tablet", "Smartwatch"),
```

```

    Category = c("Electronics", "Electronics", "Accessories", "Electronics", "Accessories")
)

reviews <- data.frame(

    ProductID = c(101, 102, 103, 104, 105),

    CustomerID = c(201, 202, 203, 204, 205),

    Rating = c(4.5, 4.2, 4.0, 4.7, 3.8)

)

write.csv(product, "Product.csv", row.names = FALSE)

write.csv(reviews, "Reviews.csv", row.names = FALSE)

product <- read.csv("Product.csv")

reviews <- read.csv("Reviews.csv")

ecommerce_data <- merge(product, reviews, by = "ProductID")

print("Merged E-Commerce Data:")

print(ecommerce_data)

rating_summary <- aggregate(

    Rating ~ Category,

    data = ecommerce_data,

    FUN = mean

)

```

```
print("Average Product Rating by Category:")

print(rating_summary)

colnames(rating_summary)[2] <- "AverageRating"

print(rating_summary)

write.csv(

  rating_summary,

  "C:/Users/adhit/Downloads/ECommerce_Rating_Report.csv",

  row.names = FALSE

)

cat("ECommerce_Rating_Report.csv created successfully!\n")
```

Output:

```

35 print(rating_summary)
36
37 write.csv(
38   rating_summary,
39   "C:/Users/adhit/Downloads/ECommerce_Rating_Report.csv",
40   row.names = FALSE
41 )
42
43
43:59 (Top Level) R Script

```

```

R 4.6.1 ~ /
2      102      Phone Electronics      202      4.2
3      103  Headphones Accessories      203      4.0
4      104      Tablet Electronics      204      4.7
5      105 Smartwatch Accessories      205      3.8
>
> rating_summary <- aggregate(
+   Rating ~ Category,
+   data = ecommerce_data,
+   FUN = mean
+ )
>
> print("Average Product Rating by Category:")
[1] "Average Product Rating by Category:"
> print(rating_summary)
  Category Rating
1 Accessories 3.900000
2 Electronics 4.466667
>
> colnames(rating_summary)[2] <- "AverageRating"
>
> print(rating_summary)
  Category AverageRating
1 Accessories      3.900000
2 Electronics      4.466667
>
> write.csv(
+   rating_summary,
+   "C:/Users/adhit/Downloads/ECommerce_Rating_Report.csv",
+   row.names = FALSE
+ )

```

6. Human Resource Management

Input:

```

employee <- data.frame(
  EmployeeID = c(101, 102, 103, 104, 105),
  Name = c("Arun", "Bala", "Charan", "Divya", "Esha"),
  Department = c("IT", "HR", "Finance", "IT", "Marketing")
)

salary <- data.frame(
  EmployeeID = c(101, 102, 103, 104, 105),
  MonthlySalary = c(45000, 50000, NA, 55000, 48000)
)

write.csv(employee, "Employee.csv", row.names = FALSE)
write.csv(salary, "Salary.csv", row.names = FALSE)

employee <- read.csv("Employee.csv")
salary <- read.csv("Salary.csv")

employee_data <- merge(employee, salary, by = "EmployeeID")

```

```
print("Merged Employee Database:")
print(employee_data)
employee_data$MonthlySalary[is.na(employee_data$MonthlySalary)] <-
  mean(employee_data$MonthlySalary, na.rm = TRUE)
employee_data$AnnualSalary <- employee_data$MonthlySalary * 12
print("Updated Employee Database:")
print(employee_data)
write.csv(
  employee_data,
  "C:/Users/adhit/Downloads/Updated_Employee_Database.csv",
  row.names = FALSE
)
cat("Updated_Employee_Database.csv created successfully!\n")
```

Output:

```

29 print(employee_data)
30
31 write.csv(
32   employee_data,
33   "C:/Users/adhit/Downloads/Updated_Employee_Database.csv",
34   row.names = FALSE
35 )
36
37 cat("Updated_Employee_Database.csv created successfully!\n")

```

37:61 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 ~/

	EmployeeID	Name	Department	MonthlySalary
1	101	Arun	IT	45000
2	102	Bala	HR	50000
3	103	Charan	Finance	NA
4	104	Divya	IT	55000
5	105	Esha	Marketing	48000

```

>
> employee_data$MonthlySalary[is.na(employee_data$MonthlySalary)] <-
+   mean(employee_data$MonthlySalary, na.rm = TRUE)
>
> employee_data$AnnualSalary <- employee_data$MonthlySalary * 12
>
> print("Updated Employee Database:")
[1] "Updated Employee Database:"
> print(employee_data)

```

	EmployeeID	Name	Department	MonthlySalary	AnnualSalary
1	101	Arun	IT	45000	540000
2	102	Bala	HR	50000	600000
3	103	Charan	Finance	49500	594000
4	104	Divya	IT	55000	660000
5	105	Esha	Marketing	48000	576000

```

>
> write.csv(
+   employee_data,
+   "C:/Users/adhit/Downloads/Updated_Employee_Database.csv"

```

7. SMART AGRICULTURE MONITORING

Input:

```
install.packages("reshape2")
```

```
library(reshape2)
```

```
agriculture <- data.frame(
```

```
  Day = c("Day1", "Day2", "Day3", "Day4", "Day5"),
```

```
  Temperature = c(30, 32, 29, 31, 28),
```

```

Humidity = c(65, 60, 70, 68, 75),
CropYield = c(120, 115, 130, 125, 135)
)
print("Original Agriculture Dataset:")
print(agriculture)

long_data <- melt(
  agriculture,
  id.vars = "Day",
  variable.name = "Parameter",
  value.name = "Value"
)
print("Long Format Dataset:")
print(long_data)
temperature <- subset(long_data, Parameter == "Temperature")
humidity <- subset(long_data, Parameter == "Humidity")
crop_yield <- subset(long_data, Parameter == "CropYield")

print("Average Temperature:")
print(mean(temperature$Value))

print("Average Humidity:")
print(mean(humidity$Value))

print("Average Crop Yield:")
print(mean(crop_yield$Value))

reconstructed_data <- dcast(
  long_data,
  Day ~ Parameter,
  value.var = "Value"
)

```

```
)
```

```
print("Reconstructed Original Dataset:")
```

```
print(reconstructed_data)
```

```
write.csv(
```

```
  reconstructed_data,
```

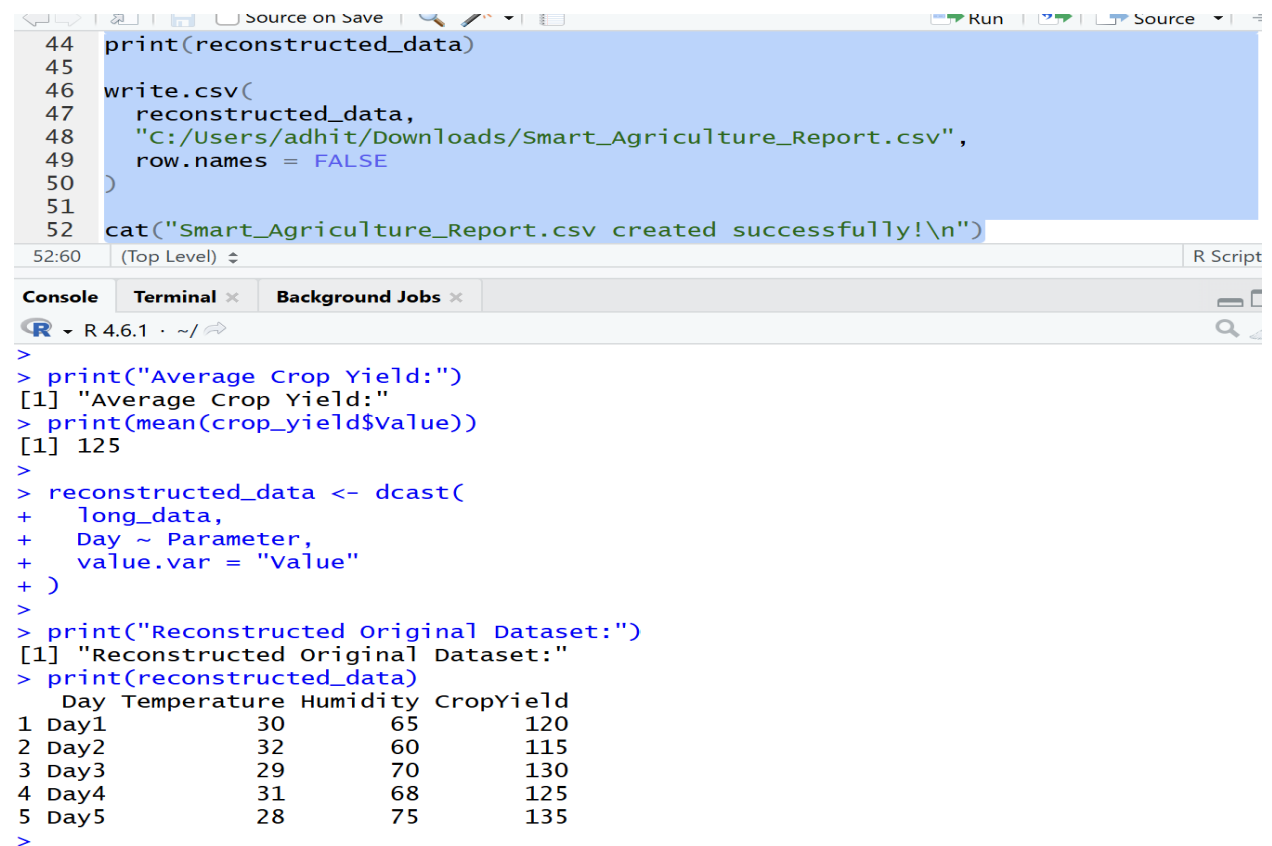
```
  "C:/Users/adhit/Downloads/Smart_Agriculture_Report.csv",
```

```
  row.names = FALSE
```

```
)
```

```
cat("Smart_Agriculture_Report.csv created successfully!\n")
```

Output:



```
44 print(reconstructed_data)
45
46 write.csv(
47   reconstructed_data,
48   "C:/Users/adhit/Downloads/Smart_Agriculture_Report.csv",
49   row.names = FALSE
50 )
51
52 cat("Smart_Agriculture_Report.csv created successfully!\n")
```

52:60 (Top Level) ↕ R Script

Console Terminal × Background Jobs ×

```
>
> print("Average Crop Yield:")
[1] "Average Crop Yield:"
> print(mean(crop_yield$Value))
[1] 125
>
> reconstructed_data <- dcast(
+   long_data,
+   Day ~ Parameter,
+   value.var = "Value"
+ )
>
> print("Reconstructed Original Dataset:")
[1] "Reconstructed Original Dataset:"
> print(reconstructed_data)
  Day Temperature Humidity CropYield
1 Day1          30        65       120
2 Day2          32        60       115
3 Day3          29        70       130
4 Day4          31        68       125
5 Day5          28        75       135
>
```

8. LOGISTICS AND SUPPLY CHAIN

Input:

```
shipment <- data.frame(  
  ShipmentID = c(101, 102, 103, 104, 105, 106),  
  Customer = c("Arun", "Bala", "Charan", "Divya", "Esha", "Farhan"),  
  City = c("Chennai", "Coimbatore", "Chennai", "Madurai", "Coimbatore", "Salem"),  
  ShipmentDate = c("2026-08-01", "2026-08-01", "2026-08-02",  
    "2026-08-02", "2026-08-03", "2026-08-03")  
)  
  
delivery <- data.frame(  
  ShipmentID = c(101, 102, 103, 104, 105, 106),  
  DeliveryStatus = c("Delivered", "Delayed", "Delivered",  
    "Delayed", "Delivered", "Delayed")  
)  
  
write.csv(shipment, "Shipment.csv", row.names = FALSE)  
write.csv(delivery, "Delivery.csv", row.names = FALSE)  
  
shipment <- read.csv("Shipment.csv")  
delivery <- read.csv("Delivery.csv")  
  
logistics_data <- merge(  
  shipment,  
  delivery,  
  by = "ShipmentID"  
)  
  
print("Merged Shipment and Delivery Data:")  
print(logistics_data)
```

```
delayed_deliveries <- subset(  
  logistics_data,  
  DeliveryStatus == "Delayed"  
)
```

```
print("Delayed Deliveries:")  
print(delayed_deliveries)
```

```
city_summary <- aggregate(  
  ShipmentID ~ City,  
  data = logistics_data,  
  FUN = length  
)
```

```
colnames(city_summary)[2] <- "TotalShipments"
```

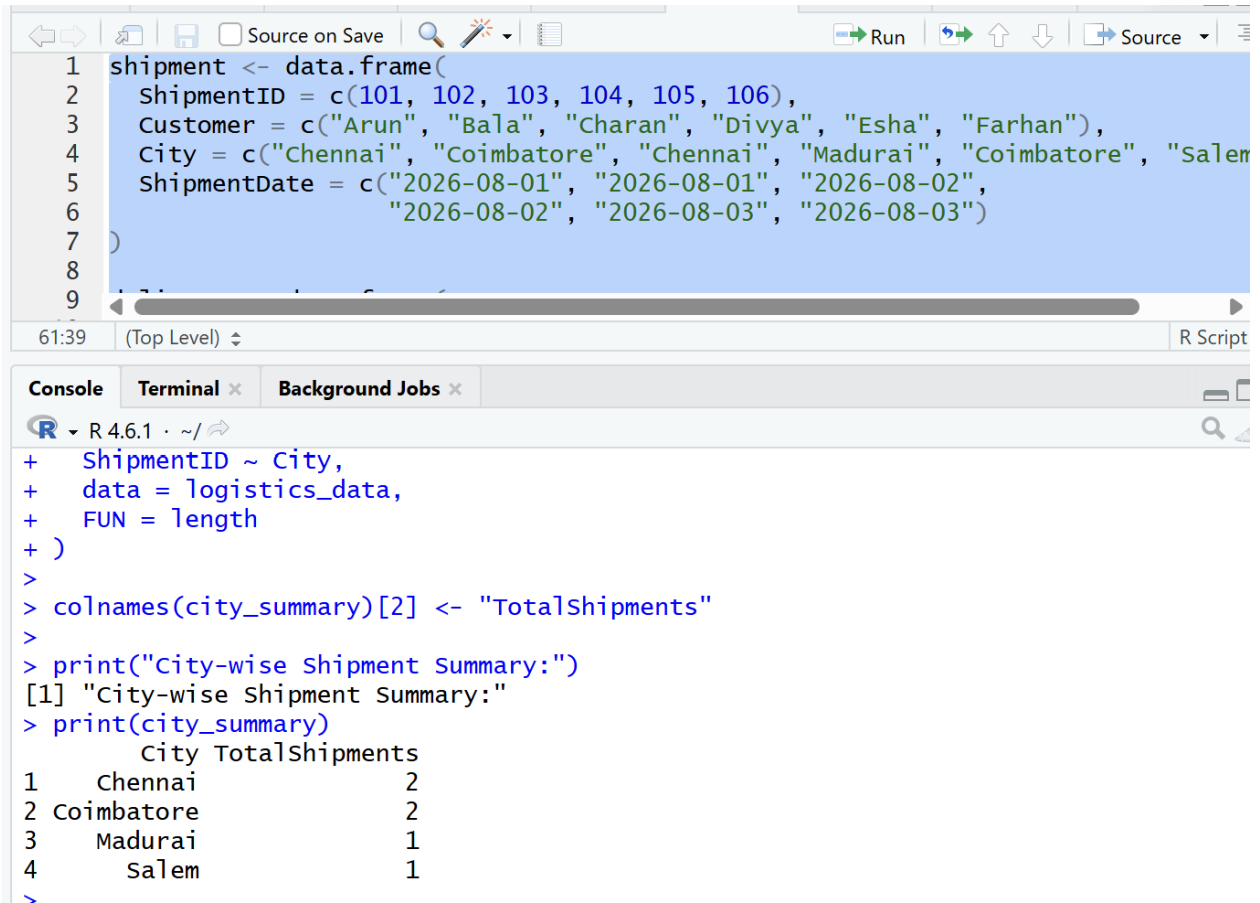
```
print("City-wise Shipment Summary:")  
print(city_summary)
```

```
write.csv(  
  city_summary,  
  "C:/Users/adhit/Downloads/City_Wise_Shipment_Summary.csv",  
  row.names = FALSE  
)
```

```
write.csv(  
  delayed_deliveries,  
  "C:/Users/adhit/Downloads/Delayed_Deliveries.csv",  
  row.names = FALSE  
)
```

```
cat("Reports created successfully!\n")
```

Output:



The screenshot shows the R Studio environment. The script editor at the top contains the following R code:

```
1 shipment <- data.frame(  
2   ShipmentID = c(101, 102, 103, 104, 105, 106),  
3   Customer = c("Arun", "Bala", "Charan", "Divya", "Esha", "Farhan"),  
4   City = c("Chennai", "Coimbatore", "Chennai", "Madurai", "Coimbatore", "Salem"),  
5   ShipmentDate = c("2026-08-01", "2026-08-01", "2026-08-02",  
6                   "2026-08-02", "2026-08-03", "2026-08-03")  
7 )  
8  
9
```

The console at the bottom shows the execution of the script:

```
+ ShipmentID ~ City,  
+ data = logistics_data,  
+ FUN = length  
+ )  
>  
> colnames(city_summary)[2] <- "TotalShipments"  
>  
> print("City-wise Shipment Summary:")  
[1] "City-wise Shipment Summary:"  
> print(city_summary)  
      City TotalShipments  
1  Chennai              2  
2 Coimbatore              2  
3  Madurai               1  
4    Salem              1  
>
```

9. UNIVERSITY STUDENT PERFORMANCE SYSTEM

Input:

```
student <- data.frame(  
  StudentID = c(101, 102, 103, 104, 105),  
  Name = c("Arun", "Bala", "Charan", "Divya", "Esha"),  
  Department = c("IT", "CSE", "ECE", "IT", "CSE")  
)
```

```

marks <- data.frame(
  StudentID = c(101, 102, 103, 104, 105),
  Semester1 = c(85, 78, 92, 75, 88),
  Semester2 = c(90, 82, 89, 80, 91),
  Semester3 = c(88, 85, 94, 78, 87)
)

write.csv(student, "Student.csv", row.names = FALSE)
write.csv(marks, "Semester_Marks.csv", row.names = FALSE)

student <- read.csv("Student.csv")
marks <- read.csv("Semester_Marks.csv")

student_data <- merge(student, marks, by = "StudentID")

student_data$AverageMarks <- rowMeans(
  student_data[, c("Semester1", "Semester2", "Semester3")]
)

print("Student Performance Report:")
print(student_data)

write.csv(
  student_data,
  "C:/Users/adhit/Downloads/Student_Performance_Report.csv",
  row.names = FALSE
)

cat("Student_Performance_Report.csv created successfully!\n")

```

Output:

```
27 print(student_data)
28
29 write.csv(
30   student_data,
31   "C:/Users/adhit/Downloads/Student_Performance_Report.csv",
32   row.names = FALSE
33 )
34
35 cat("Student_Performance_Report.csv created successfully!\n")
```

35:62 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 ~/

```
+ )
>
> marks <- data.frame(
+   StudentID = c(101, 102, 103, 104, 105),
+   Semester1 = c(85, 78, 92, 75, 88),
+   Semester2 = c(90, 82, 89, 80, 91),
+   Semester3 = c(88, 85, 94, 78, 87)
+ )
>
> write.csv(student, "Student.csv", row.names = FALSE)
> write.csv(marks, "Semester_Marks.csv", row.names = FALSE)
>
> student <- read.csv("Student.csv")
> marks <- read.csv("Semester_Marks.csv")
>
> student_data <- merge(student, marks, by = "StudentID")
>
> student_data$AverageMarks <- rowMeans(
+   student_data[, c("Semester1", "Semester2", "Semester3")]
+ )
>
> print("Student Performance Report:")
[1] "Student Performance Report:"
> print(student_data)
```

	StudentID	Name	Department	Semester1	Semester2	Semester3	AverageMarks
1	101	Arun	IT	85	90	88	87.66667
2	102	Bala	CSE	78	82	85	81.66667
3	103	Charan	ECE	92	89	94	91.66667
4	104	Divya	IT	75	80	78	77.66667
5	105	Esha	CSE	88	91	87	88.66667

10.INSURANCE CLAIM PROCESSING

Input:

```
policy <- data.frame(
```

```

PolicyID = c(101, 102, 103, 104, 105),
Customer = c("Arun", "Bala", "Charan", "Divya", "Esha"),
PolicyType = c("Health", "Life", "Health", "Vehicle", "Life")
)

claims <- data.frame(
  ClaimID = c(1, 2, 3, 4, 5, 6, 7),
  PolicyID = c(101, 101, 102, 103, 103, 103, 104),
  ClaimAmount = c(50000, 30000, 80000, 45000, 25000, 35000, 60000)
)

write.csv(policy, "Policy.csv", row.names = FALSE)
write.csv(claims, "Claims.csv", row.names = FALSE)

policy <- read.csv("Policy.csv")
claims <- read.csv("Claims.csv")

insurance_data <- merge(policy, claims, by = "PolicyID")

print("Merged Insurance Claim Data:")
print(insurance_data)

claim_count <- aggregate(
  ClaimID ~ Customer,
  data = insurance_data,
  FUN = length
)

multiple_claims <- subset(claim_count, ClaimID > 1)

print("Customers with Multiple Claims:")

```

```
print(multiple_claims)
```

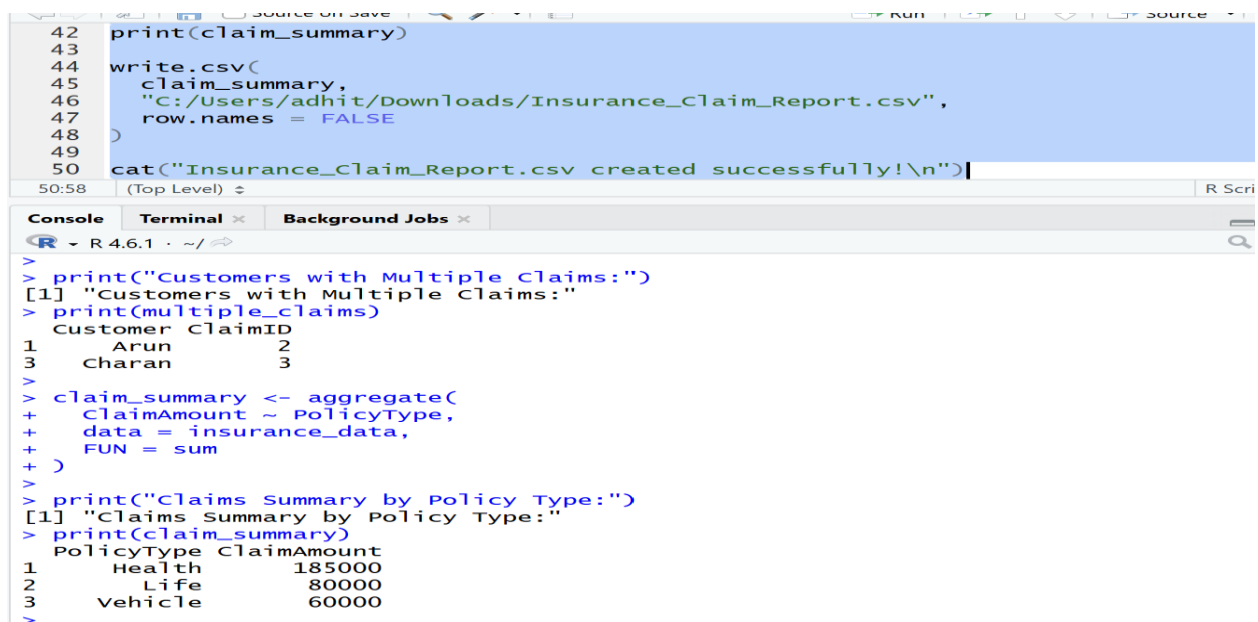
```
claim_summary <- aggregate(  
  ClaimAmount ~ PolicyType,  
  data = insurance_data,  
  FUN = sum  
)
```

```
print("Claims Summary by Policy Type:")  
print(claim_summary)
```

```
write.csv(  
  claim_summary,  
  "C:/Users/adhit/Downloads/Insurance_Claim_Report.csv",  
  row.names = FALSE  
)
```

```
cat("Insurance_Claim_Report.csv created successfully!\n")
```

Output:



The screenshot shows an R Studio console window with the following code and output:

```
> print(claim_summary)
[1] "Customers with Multiple Claims:"
> print(multiple_claims)
  Customer ClaimID
1     Arun        2
3    Charan        3
> 
> claim_summary <- aggregate(
+   ClaimAmount ~ PolicyType,
+   data = insurance_data,
+   FUN = sum
+ )
> 
> print("Claims Summary by Policy Type:")
[1] "Claims Summary by Policy Type:"
> print(claim_summary)
  PolicyType ClaimAmount
1    Health      185000
2     Life       80000
3   Vehicle       60000
>
```

The console output displays the results of the R code execution, including the print statement for multiple claims, the aggregation of claim amounts by policy type, and the final print statement for the claims summary by policy type.

11.TELECOMMUNICATIONS NETWORK MONITORING

Input:

```
region1 <- data.frame(  
  Region = "North",  
  Date = c("2026-08-01", "2026-08-02", "2026-08-03"),  
  DownloadSpeed = c(85, 90, 88),  
  UploadSpeed = c(40, 45, 42)  
)
```

```
region2 <- data.frame(  
  Region = "South",  
  Date = c("2026-08-01", "2026-08-02", "2026-08-03"),  
  DownloadSpeed = c(95, 98, 92),  
  UploadSpeed = c(50, 52, 48)  
)
```

```
region3 <- data.frame(  
  Region = "East",  
  Date = c("2026-08-01", "2026-08-02", "2026-08-03"),  
  DownloadSpeed = c(75, 80, 78),  
  UploadSpeed = c(35, 38, 36)  
)
```

```
write.csv(region1, "Network_North.csv", row.names = FALSE)  
write.csv(region2, "Network_South.csv", row.names = FALSE)  
write.csv(region3, "Network_East.csv", row.names = FALSE)
```

```
region1 <- read.csv("Network_North.csv")  
region2 <- read.csv("Network_South.csv")  
region3 <- read.csv("Network_East.csv")
```

```
network_data <- rbind(region1, region2, region3)
```

```
print("Combined Network Data:")
```

```
print(network_data)
```

```
download_summary <- aggregate(
```

```
  DownloadSpeed ~ Region,
```

```
  data = network_data,
```

```
  FUN = mean
```

```
)
```

```
print("Average Download Speed by Region:")
```

```
print(download_summary)
```

```
write.csv(
```

```
  download_summary,
```

```
  "C:/Users/adhit/Downloads/Network_Performance_Report.csv",
```

```
  row.names = FALSE
```

```
)
```

```
cat("Network_Performance_Report.csv created successfully!\n")
```

Output:

```
42 print(download_summary)
43
44 write.csv(
45   download_summary,
46   "C:/Users/adhit/Downloads/Network_Performance_Report.csv",
47   row.names = FALSE
48 )
49
50 cat("Network_Performance_Report.csv created successfully!\n")
```

50:62 (Top Level) R Script

Console Terminal Background Jobs

```
> region1 <- read.csv("Network_East.csv")
>
> network_data <- rbind(region1, region2, region3)
>
> print("Combined Network Data:")
[1] "Combined Network Data:"
> print(network_data)
  Region      Date DownloadSpeed UploadSpeed
1 North 2026-08-01           85           40
2 North 2026-08-02           90           45
3 North 2026-08-03           88           42
4 South 2026-08-01           95           50
5 South 2026-08-02           98           52
6 South 2026-08-03           92           48
7 East  2026-08-01           75           35
8 East  2026-08-02           80           38
9 East  2026-08-03           78           36
>
> download_summary <- aggregate(
+   DownloadSpeed ~ Region,
+   data = network_data,
+   FUN = mean
+ )
>
> print("Average Download Speed by Region:")
[1] "Average Download Speed by Region:"
> print(download_summary)
  Region DownloadSpeed
1 East      77.66667
2 North     87.66667
3 South     95.00000
```

12.ENERGY CONSUMPTION ANALYSIS

Input:

```
library(reshape2)
```

```
energy <- data.frame(
  Date = c("2026-08-01", "2026-08-02", "2026-08-03"),
  Chennai = c(450, 480, 500),
  Coimbatore = c(350, 370, 390),
  Madurai = c(300, 320, 340),
  Salem = c(280, 300, 310)
)
```

```
write.csv(energy, "Energy.csv", row.names = FALSE)
```

```
energy <- read.csv("Energy.csv")
```

```
print("Original Energy Dataset:")
```

```
print(energy)
```

```
long_energy <- melt(  
  energy,  
  id.vars = "Date",  
  variable.name = "Region",  
  value.name = "Consumption"  
)
```

```
print("Long Format Energy Dataset:")
```

```
print(long_energy)
```

```
daily_consumption <- dcast(  
  long_energy,  
  Date ~ Region,  
  value.var = "Consumption",  
  fun.aggregate = sum  
)
```

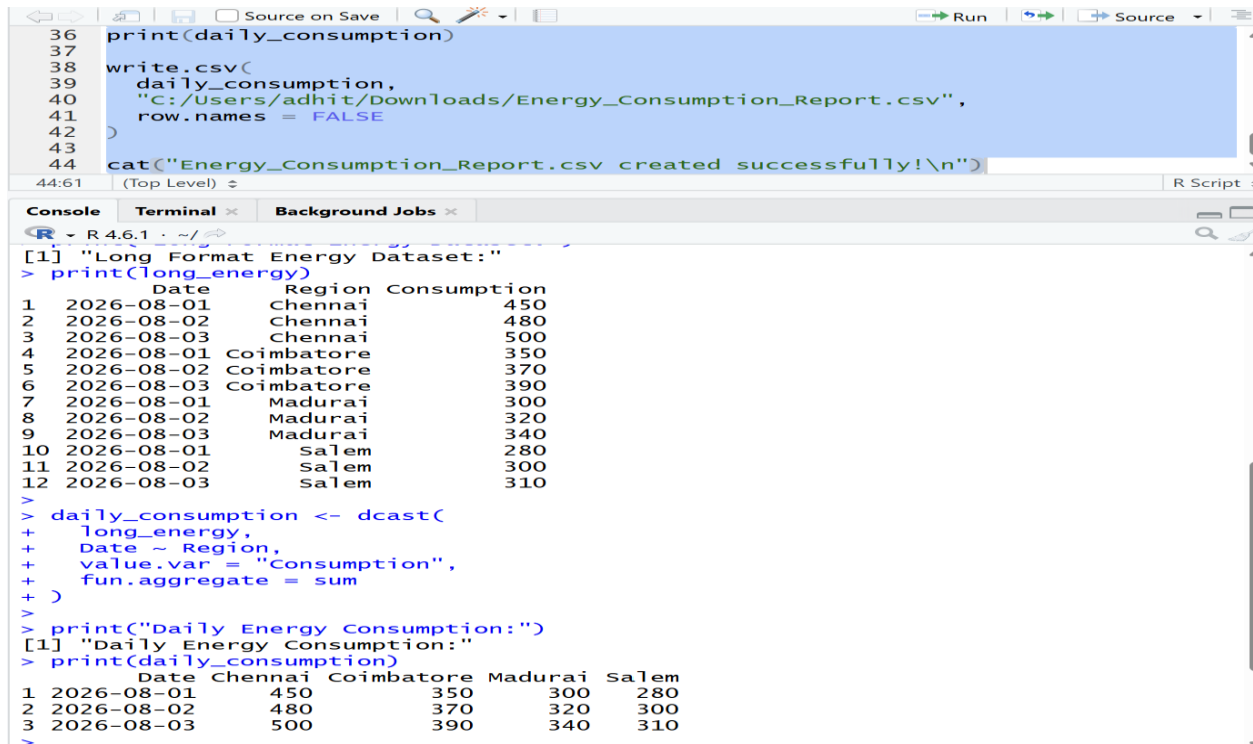
```
print("Daily Energy Consumption:")
```

```
print(daily_consumption)
```

```
write.csv(  
  daily_consumption,  
  "C:/Users/adhit/Downloads/Energy_Consumption_Report.csv",  
  row.names = FALSE  
)
```

```
cat("Energy_Consumption_Report.csv created successfully!\n")
```

Output:



The screenshot shows the R Studio environment. The script editor at the top contains the following code:

```
36 print(daily_consumption)
37
38 write.csv(
39   daily_consumption,
40   "C:/Users/adhit/Downloads/Energy_Consumption_Report.csv",
41   row.names = FALSE
42 )
43
44 cat("Energy_Consumption_Report.csv created successfully!\n")
```

The console at the bottom shows the output of the code execution:

```
[1] "Long Format Energy Dataset:"
> print(long_energy)
  Date      Region Consumption
1 2026-08-01 Chennai         450
2 2026-08-02 Chennai         480
3 2026-08-03 Chennai         500
4 2026-08-01 Coimbatore       350
5 2026-08-02 Coimbatore       370
6 2026-08-03 Coimbatore       390
7 2026-08-01 Madurai          300
8 2026-08-02 Madurai          320
9 2026-08-03 Madurai          340
10 2026-08-01 Salem           280
11 2026-08-02 Salem           300
12 2026-08-03 Salem           310
>
> daily_consumption <- dcast(
+   long_energy,
+   Date ~ Region,
+   value.var = "Consumption",
+   fun.aggregate = sum
+ )
>
> print("Daily Energy Consumption:")
[1] "Daily Energy Consumption:"
> print(daily_consumption)
  Date Chennai Coimbatore Madurai Salem
1 2026-08-01    450        350    300    280
2 2026-08-02    480        370    320    300
3 2026-08-03    500        390    340    310
>
```

13. AIRLINE RESERVATION SYSTEM

Input:

```
passenger <- data.frame(
  BookingID = c(101, 102, 103, 104, 105),
  PassengerName = c("Arun", "Bala", "Charan", "Divya", "Esha"),
  FlightNumber = c("AI101", "AI102", "AI103", "AI104", "AI105")
)
```

```
flight <- data.frame(
  FlightNumber = c("AI101", "AI102", "AI103", "AI104", "AI105"),
  Source = c("Chennai", "Delhi", "Mumbai", "Bangalore", "Kochi"),
```

```
Destination = c("Delhi", "Mumbai", "Chennai", "Kochi", "Chennai"),  
FlightStatus = c("On Time", "Delayed", "On Time", "Delayed", NA)  
)
```

```
write.csv(passenger, "Passenger.csv", row.names = FALSE)  
write.csv(flight, "Flight.csv", row.names = FALSE)
```

```
passenger <- read.csv("Passenger.csv")  
flight <- read.csv("Flight.csv")
```

```
airline_data <- merge(  
  passenger,  
  flight,  
  by = "FlightNumber"  
)
```

```
airline_data$FlightStatus[  
  is.na(airline_data$FlightStatus)  
] <- "Unknown"
```

```
print("Final Airline Reservation Report:")  
print(airline_data)
```

```
delayed_flights <- subset(  
  airline_data,  
  FlightStatus == "Delayed"  
)
```

```
print("Delayed Flights:")  
print(delayed_flights)
```

```

write.csv(
  airline_data,
  "C:/Users/adhit/Downloads/Airline_Reservation_Report.csv",
  row.names = FALSE
)

```

```

cat("Airline_Reservation_Report.csv created successfully!\n")

```

Output:

```

39 print(delayed_flights)
40
41 write.csv(
42   airline_data,
43   "C:/Users/adhit/Downloads/Airline_Reservation_Report.csv",
44   row.names = FALSE
45 )
46
47 cat("Airline_Reservation_Report.csv created successfully!\n")

```

47:62 (Top Level) R St

Console Terminal Background Jobs

R 4.6.1 ~/ ↗

```

+ by = "FlightNumber"
+ )
>
> airline_data$FlightStatus[
+   is.na(airline_data$FlightStatus)
+ ] <- "Unknown"
>
> print("Final Airline Reservation Report:")
[1] "Final Airline Reservation Report:"
> print(airline_data)
  FlightNumber BookingID PassengerName Source Destination FlightStatus
1          AI101       101         Arun  Chennai         Delhi    On Time
2          AI102       102          Bala   Delhi         Mumbai   Delayed
3          AI103       103        Charan  Mumbai         Chennai  On Time
4          AI104       104        Divya Bangalore         Kochi   Delayed
5          AI105       105          Esha   Kochi         Chennai Unknown
>
> delayed_flights <- subset(
+   airline_data,
+   FlightStatus == "Delayed"
+ )
>
> print("Delayed Flights:")
[1] "Delayed Flights:"
> print(delayed_flights)
  FlightNumber BookingID PassengerName Source Destination FlightStatus
2          AI102       102          Bala   Delhi         Mumbai   Delayed
4          AI104       104        Divya Bangalore         Kochi   Delayed
>

```

14.FINANCIAL MARKET ANALYSIS

Input:

```
company <- data.frame(  
  CompanyID = c(101, 102, 103),  
  Company = c("ABC Ltd", "XYZ Ltd", "PQR Ltd"),  
  Sector = c("Technology", "Banking", "Automobile")  
)  
  
stock <- data.frame(  
  CompanyID = c(101, 101, 101, 102, 102, 102, 103, 103, 103),  
  Month = c("January", "February", "March",  
    "January", "February", "March",  
    "January", "February", "March"),  
  StockPrice = c(150, 160, 170, 200, 210, 220, 100, 110, 115)  
)  
  
write.csv(company, "Company.csv", row.names = FALSE)  
write.csv(stock, "Stock.csv", row.names = FALSE)  
  
company <- read.csv("Company.csv")  
stock <- read.csv("Stock.csv")  
  
financial_data <- merge(  
  company,  
  stock,  
  by = "CompanyID"  
)  
  
print("Merged Financial Data:")  
print(financial_data)
```

```
monthly_performance <- aggregate(  
  StockPrice ~ Company + Month,  
  data = financial_data,  
  FUN = mean  
)  
  
print("Company-wise Monthly Stock Performance:")  
print(monthly_performance)  
  
performance_report <- reshape(  
  monthly_performance,  
  idvar = "Company",  
  timevar = "Month",  
  direction = "wide"  
)  
  
print("Reshaped Stock Performance:")  
print(performance_report)  
  
write.csv(  
  performance_report,  
  "C:/Users/adhit/Downloads/Financial_Market_Report.csv",  
  row.names = FALSE  
)  
  
cat("Financial_Market_Report.csv created successfully!\n")
```

Output:

```
47 print(performance_report)
48
49 write.csv(
50   performance_report,
51   "C:/Users/adhit/Downloads/Financial_Market_Report.csv",
52   row.names = FALSE
53 )
54
55 cat("Financial_Market_Report.csv created successfully!\n")
```

55:59 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 ~/

```
+ )
>
> print("Company-wise Monthly Stock Performance:")
[1] "Company-wise Monthly Stock Performance:"
> print(monthly_performance)
  Company      Month StockPrice
1 ABC Ltd February         160
2 PQR Ltd February         110
3 XYZ Ltd February         210
4 ABC Ltd January          150
5 PQR Ltd January          100
6 XYZ Ltd January          200
7 ABC Ltd March            170
8 PQR Ltd March            115
9 XYZ Ltd March            220
>
> performance_report <- reshape(
+   monthly_performance,
+   idvar = "Company",
+   timevar = "Month",
+   direction = "wide"
+ )
>
> print("Reshaped Stock Performance:")
[1] "Reshaped Stock Performance:"
> print(performance_report)
  Company StockPrice.February StockPrice.January StockPrice.March
1 ABC Ltd                160                150                170
2 PQR Ltd                110                100                115
3 XYZ Ltd                210                200                220
```

15.SMART CITY TRAFFIC MANAGEMENT

Input:

```
intersection1 <- data.frame(
  Location = "Anna Nagar",
  Hour = c(8, 10, 12, 14, 18),
```

```

    TrafficVolume = c(450, 300, 350, 320, 600)
)

intersection2 <- data.frame(
  Location = "T Nagar",
  Hour = c(8, 10, 12, 14, 18),
  TrafficVolume = c(500, 350, 400, 380, 700)
)

intersection3 <- data.frame(
  Location = "Guindy",
  Hour = c(8, 10, 12, 14, 18),
  TrafficVolume = c(550, 400, 450, 420, 750)
)

write.csv(intersection1, "Traffic_Anna_Nagar.csv", row.names = FALSE)
write.csv(intersection2, "Traffic_T_Nagar.csv", row.names = FALSE)
write.csv(intersection3, "Traffic_Guindy.csv", row.names = FALSE)

intersection1 <- read.csv("Traffic_Anna_Nagar.csv")
intersection2 <- read.csv("Traffic_T_Nagar.csv")
intersection3 <- read.csv("Traffic_Guindy.csv")

traffic_data <- rbind(
  intersection1,
  intersection2,
  intersection3
)

print("Combined Traffic Data:")
print(traffic_data)

```

```
peak_traffic <- aggregate(  
  TrafficVolume ~ Location,  
  data = traffic_data,  
  FUN = max  
)
```

```
print("Peak-Hour Traffic by Location:")  
print(peak_traffic)
```

```
peak_details <- traffic_data[  
  ave(  
    traffic_data$TrafficVolume,  
    traffic_data$Location,  
    FUN = function(x) x == max(x)  
  ) == TRUE,  
]
```

```
print("Peak Traffic Details:")  
print(peak_details)
```

```
write.csv(  
  peak_details,  
  "C:/Users/adhit/Downloads/Smart_City_Traffic_Report.csv",  
  row.names = FALSE  
)
```

```
cat("Smart_City_Traffic_Report.csv created successfully!\n")
```

Output:

```
54 print(peak_details)
55
56 write.csv(
57   peak_details,
58   "C:/Users/adhit/Downloads/Smart_City_Traffic_Report.csv",
59   row.names = FALSE
60 )
61
62 cat("Smart_City_Traffic_Report.csv created successfully!\n")
```

62:61 (Top Level) ↕

Console Terminal × Background Jobs ×

R 4.6.1 · ~/

```
>
> peak_traffic <- aggregate(
+   TrafficVolume ~ Location,
+   data = traffic_data,
+   FUN = max
+ )
>
> print("Peak-Hour Traffic by Location:")
[1] "Peak-Hour Traffic by Location:"
> print(peak_traffic)
      Location TrafficVolume
1 Anna Nagar           600
2   Guindy             750
3   T Nagar             700
>
> peak_details <- traffic_data[
+   ave(
+     traffic_data$TrafficVolume,
+     traffic_data$Location,
+     FUN = function(x) x == max(x)
+   ) == TRUE,
+ ]
>
> print("Peak Traffic Details:")
[1] "Peak Traffic Details:"
> print(peak_details)
      Location Hour TrafficVolume
5 Anna Nagar    18           600
10  T Nagar     18           700
15  Guindy      18           750
```